CHAPTER TWO:

BASIC CONCEPTS OF LOGIC :

Arguments, Premises and Conclusions

What is Logic?

- Branches of Philosophy
- **Science of Argument**/Skillful **Reasoning**
- Study of Methods/Principles
- A basic **tool** of Philosophers
- A central critical thinking tool

Logic…..

- is generally be defined as a **philosophical science** that **evaluates arguments**.
- The primary aim of logic is **to develop** a system of methods and principles that we may use as **criteria for evaluating the** **<u>arguments of others</u>** and as **guides in constructing** **arguments of our own.**

Components of Logic

- **Argument** is a group of statements (TV) that claims to prove something.
- **Premise** is a set of statement that setforth reasons or evidences to believe in the other.
- **Conclusion**: A statement derived from the given premise.
- **Argument** is a claim supported by other claims.

Cont…

- The argument can be **Thought, discourse dialectic** or any **Message** that is being conveyed either through speech, writing, performance, or other media.

- An argument, in philosophy, **isn't** just a **shouting match**; **It is about maintaining once own beliefs on the basis of reason**.

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Purpose of Studying Logic?

- It helps us to **develop the skill needed to construct sound (good**) and **fallacy-free**
- Help us to **construct arguments of one's own** and **to evaluate the arguments of others;**
- It helps us **to distinguish good arguments from bad arguments;**
- It helps us to **understand** and **identify** the **common logical errors in reasoning**;

Cont…

- It helps us to understand and identify the common confusions that often happen due to misuse of language;
- It **enables us to disclose ill-conceived policies** in the **political sphere**, to be careful of disguises, and to distinguish the rational from irrational and the sane from the insane and so on.

What is an Argument?

- it is a **group of two** or **more statements**, **one or more of which** (the **premise**) are **claimed to provide support** for, or reason to believe, one of the other, the (**conclusion**).
- A **statement** is a **declarative sentence** that has a **truth-value** of either true or false.

Cont….

- **Propositions** are the **building blocks** of our reasoning
- A proposition **asserts** that **something is the case.**
- We may affirm a proposition, or deny it- but every proposition either asserts what really is the case, or it does not.
- Therefore, **every proposition** is either **true** or it is **false**.

Cont….

- **Example**

a). Dr. Abiy Ahmed the current Prime Minister of Ethiopia.

b) Mekelle is the capital city of Tigray Region.

c) Ethiopia was colonized by Germany.

Sentences which are not statements include:

- ✓ Questions
- ✓ Commands
- ✓ Proposals
- ✓ Suggestion and
- ✓ Exclamation

❖ **Example:**

a). **Would you close the window? (Question)**
b) Let us study together. (Proposal)
c) Right on! (Exclamation)
d) I suggest that you read philosophy texts. (Suggestion)
e) Give me your ID Card, Now! (Command)

Examples of Arguments

1. All animals are mammals
   <u>Cows are animals</u>
   So, cows are mammals.

2. All men are Ethiopians
   <u>Obama is man</u>
   Thus, Obama is Ethiopians.

#Good Argument

Cont….

3. ***Some men are Ethiopians***

Obama is man

∴, Obama is Ethiopian.

#Bad argument

- ***All Indians have long hair***

Deepak is Indian

∴, Deepak has Long hair.

Good Argument vs Bad Argument

- An argument is **good** if the premises **really supports** the conclusion
- An argument is **bad** if the premises in **not sufficiently support** the conclusion

- **Examples**

1. Some men are tall

Dawit is man

Therefore, Dawit is tall

Cont….

2. All Indians have longhair

Deepak is Indian

So, Deepak has long hair.

3. All Indians have long hair

___Chaltu has long hair_

So, Chaltu is Indian

Argument vs Non-Argument

- To be an argument, the passage must claim to prove something.
- Proving something is followed by 2 claims:

1.Factual Claim

2.Inferential Claim

- **Thus, An argument is supported by claims.**

Non-argument_

- Non- inferential claim
- Lacks reasoning process
- Lacks inferences
- No intention **to prove**
- It is an opinion

- ***An opinion that is not supported by claim***

Simple vs Complex
Non-argumentative claims

❖ **Simple Non- IC are**:

✓ Warning, Advice, Report ,Opinion/belief, and Loose statement

❖ **Complex Non-IC are:**

- ✓ **Illustration**=Example
- ✓ **Explanation**=matter of facts/ shed lights on some accepted facts
- ✓ **Expository = Elaboration**
- ✓ **Conditional statement=**
  If…then …

1. Passages Lacking Inferential Claims

- **Missing a claim** that a reasoning process is being expressed- that **potential premises support a conclusion** or that a potential conclusion follows from premises. These are:
  - ✓ **Warnings**
  - ✓ **Pieces of Advice**
  - ✓ **Statements of Belief/opinion**

Cont..

Example:

- I think a nation such as ours, with its moral traditions and commitments has a further responsibility to know how we became drawn into this conflict, and to learn the lessons it has to teach us for future.

Loosely Associated Statements

- may be about the same general subject, but they **lack a claim that one of them is proved by the others;**
- **Example:**
  - **Not to honor men of worth will keep the people from contention; not to value goods that are hard to come by will keep them from theft; not to display what is desirable will keep them from being unsettled of mind**.

❑ A Report

- consists of a group of **statements that convey information about some situation or event.**

❑ Example:

- **A powerful car bomb blew up outside the regional telephone company headquarters in Medellin, injuring 25 people and causing millions of dollars of damage to nearby buildings, police said. A police statement said the 198- pound bomb was packed into a milk churn hidden in the back of a stolen car.**

An Expository Passage

- is a kind of discourse that **begins with a topic sentence followed by one or more sentences that develop the topic sentence**.
- If the objective is **not to prove the topic sentence** but only to expand it or elaborate it, then **there is no argument.**

- **Example**:

- There is a stylized relation of artist to mass audience in the sports, especially in baseball. Each player develops a style of his own- the swagger as he steps to the plate, the unique windup a pitcher has the clean swinging and hard-driving hits, the precision quickness and grace of infield and outfield, the sense of surplus power behind whatever is done.

An Illustration

- consists of a **statement** about a certain subject combined **with a reference to one or more specific instances intended to exemplify** that statement.

Note: Illustrations are often confused with arguments b/c many of them contain indicator words such as “thus”

- **Example:**

✓ Chemical elements, as well as compounds, can be represented by molecular formulas. Thus, oxygen is represented by “O2” sodium chloride by “NaC1”, and sulfuric acid by “H2So4”

✓ Mammals are vertebrate animals that nourish their young with milk. For example, cats, horses, goats, monkeys, and humans are mammals.

Conditional statements

- A conditional statement is an **"If… then… " Statement.**
- Every conditional statement is made up of **two parts.**

  a). **Antecedent- comes after the word "if"**

  b). **Consequent- comes after the word "then"**

- In a conditional statement, there is **no claim** that either the antecedent or the consequent presents **evidence**
- In other words, there is **no assertion** that either the antecedent or the consequent is true.

Relation b/n conditional statements and arguments

- The relation b/n conditional statements and arguments may now be **summarized** as follows:
  - **A single conditional statement is not an argument.**
  - **A conditional statement may serve as either the premise or the conclusion (or both) of an argument**
  - **The inferential content of a conditional statement may be re-expressed to form an argument**

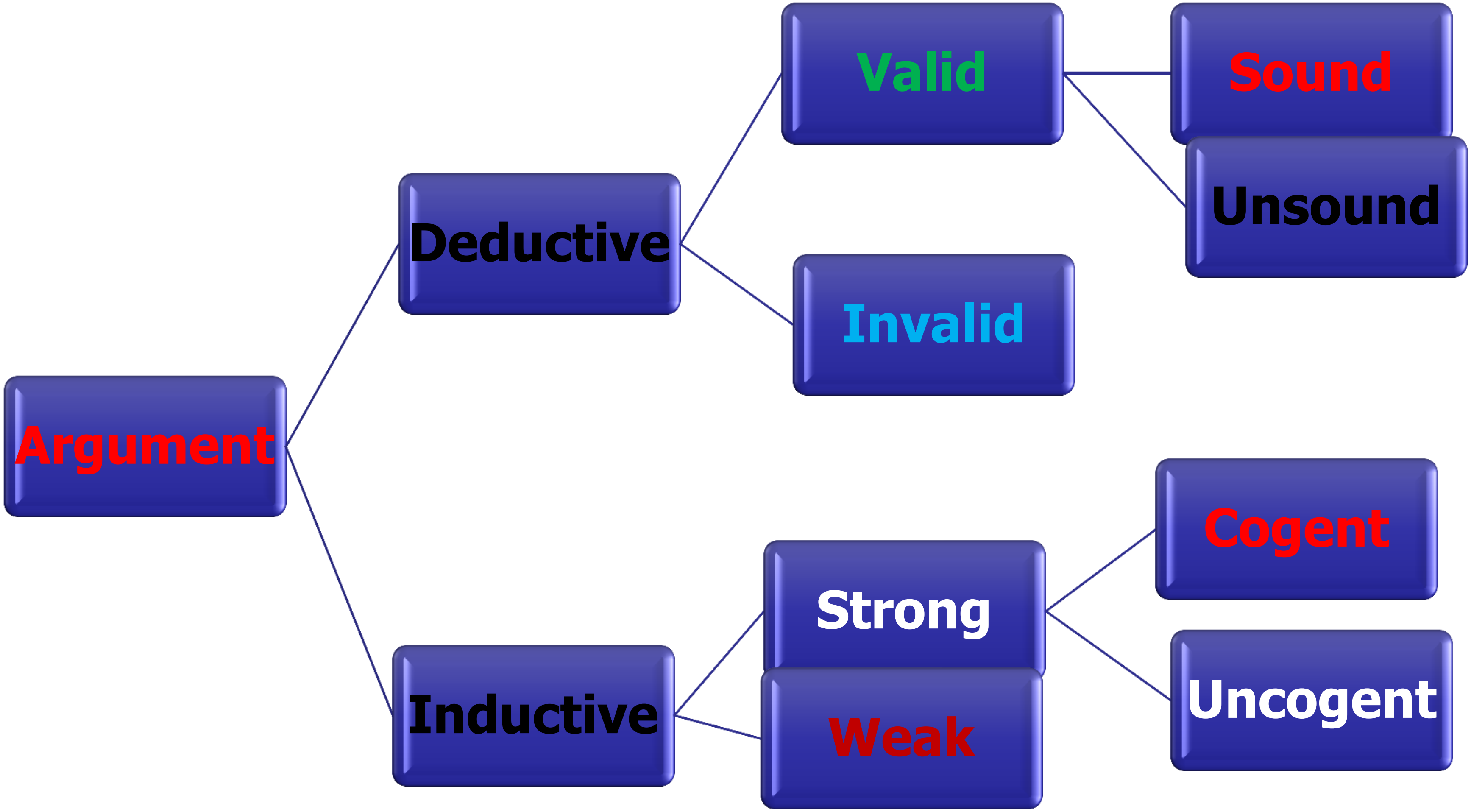
Explanations:

- an explanation consists of a statement or group of statements **intended to shed light on some phenomenon that is usually accepted as a matter of fact.**
- **Example:**
- **Almaz got sick because she ate too much**
- **It is already accepted that Almaz got sick and the question is, "why did she get sick?" "What caused her sickness?"**

Cont..

- Explanations are sometimes **mistaken for arguments** b/c they often **contain the indicator word** "**because**".
- **Yet explanations are not arguments for the ff reason:**
  - **explanans is intended to show why something is the case, whereas in an argument the premises are intended to prove that something is the case.**

Type of Argument



There are two kind of argument

- **i). Deductive (**Truth/ Proof)…**law of Necessity**
- **ii). Inductive (**Belief/Probability)…**law of Probability**
- **DA…**is an argument such that **if PT**, then **C cannot be F**.
  - If **Pt**, the **C never be false**.
- **C** is claimed to follow **Necessarily/absolutely** from the premises

Cont….

- **IA**…is an argument such that **if PT**, then **its is improbable for the C to be True.**
- **C** is claim to follow **only probably from the premises**

Examples

- ❖ 1. All men are tigers

  Samy is man____

  So, Samy is tiger.

- ❖2. Some numbers are odd

  Three is number____

  So, three is odd.

Cont…

- 3. Some women are tall

Sara is women

So, Sara is tall

- 4. A vast majority of JU are Indians.

Dawit is JU stuff.

So, Dawit is Indian

How to Identify Deductive from Inductive Argument?

- Identifying DA from IA involves 4 steps as follow:

i). Look for the occurrence of special indicator words to identify DA from IA

- **Indicator words for DA** include: "**necessarily,**" "**certainly,**" "**absolutely,**" **and** "**definitely**."

- **Indicator words for IA** include: "**probably,** "**improbable,**" "**plausible,**" "**implausible,**" "**likely,**" "**unlikely,**" **and** "**reasonable to conclude**."

Cont…

ii). Look at the actual strength of the inferential link between premises and conclusion.

- Do the Premise **necessarily** or **Probably** support the Conclusion?
- **If** the connection is supposed to be **necessary**, then the **argument is a deductive argument.**
- **If** the conclusion is **just probably true**, given the truth of the premises, the **argument is inductive**

Cont….

iii). The character or form of argumentation.

- This refers to various forms of DA and IA
- ***There are some* branches of Deductive Logic:**

➢ **Mathematics**

➢ **Analytic definition** (involves exactly same meaning)

➢ **Syllogism** (**Categorical**, **Hypothetical & Disjunctive**)

***There are also some* branches of Inductive Logic**

➢ **Prediction, Analogy, Generalization, Causal effects, Argument from Authority, sign or symbol**

Branches of Deductive Logic

1. Mathematics: (except Statistics)

- **It** is an argument in which the **conclusion depends on some purely arithmetic or geometric computation or measurement.**

- **Example**

- **A shopper puts two oranges into a bag that already contains three pieces of fruit and conclude that the bag now contains five pieces of fruit.**

Analytic Definition

- **It** is an argument in which the **conclusion** is claimed to **depend merely upon the definition of some word or phrase** used in the **premise or conclusion**.
- For **example,** one might argue that:

- ✓ Because Alex is **mendacious(dishonest)**, it follows that he tells **lies.**
- ✓ God is **all-powerful**, it follows that God **is omnipotent**

Syllogism
(founded by Aristotle)

- It is an argument consisting of **exactly two** premises and **one conclusion**.

➢ **Syllogism = 2P + 1 C**

- **There are 3 kind of Syllogisms.**

1. Categorical Syllogism

- **It** is a syllogism in which each **statement begins** with one of the words “**all,**” “**no,**” **or** “**some.**”

Example:

- **All ancient forests are sources of wonder.**

Some ancient forests are targets of the timber industry.

Therefore, some sources of wonder are targets of the timber industry.

2. Hypothetical Syllogism

- **It** is a syllogism **having a conditional statement** for one or both of its premises.
  - **Examples:**
- **If Helen study hard, she will get A.**

If she gets A, then she will get scholarship.

Therefore, if Helen study hard, she will get scholarship.

- **If there is all loving God, then there is no pain and death.**

There is pain and death

Therefore, if there is no all loving God.

3. Disjunctive Syllogism

- **DS** is a syllogism having a disjunctive statement (i.e., an "**either. . . or** . . .") for one of its premises and one it should be eliminated to form valid argument.

Example:

- **Either Tomas is studying philosophy or he is studying physics.**
- **Tomas is studying physics.**
- **Therefore, Tomas is not studying philosophy.**

Inductive argument forms:

1. Prediction

- A prediction is an argument that **proceeds from our knowledge of the past to a claim about the future**.
- **Example:**

a). Someone might argue that because certain meteorological phenomena have been observed to develop over a certain region of central Missouri, a storm will occur there in six hours.

b). **The sun has been rising so far. Therefore, the sun will rise tomorrow**.

2. Analogy

- **is** an argument that depends on the existence of an analogy, or **similarity, between two things** or states of affairs.
- **Example:**
- **Someone might argue that because Mr X's Samsung mobile is so active, it follows that Mr Y's Samsung mobile must also be active**.

3. Generalization

- A generalization is an **argument that proceeds from the knowledge of a selected sample** to some **claim** about the **whole group**.

- **Example:**

One might argue that because three apples selected from a certain crate were especially tasty and juicy, all the apples from that crate are especially tasty and juicy.

4. Argument from Authority

- An argument from authority is an argument that **concludes something is true** because **a presumed expert or witness has said that it is**.

  - **Example:**

- **A person might argue that earnings for *HP* Corporation will be up in the coming quarter because of a statement to that effect by an investment counselor.**

5. Argument based on Sign/Symbol

- An argument that **proceeds from the knowledge** of a **certain sign to a knowledge** of the thing or situation that **the sign symbolizes**.
  - **Example:**

- **When driving on an unfamiliar highway one might see a sign indicating that the road makes several sharp turns one mile ahead. Based on this information, one might argue that the road does indeed make several sharp turns one mile ahead.**

6. Causal Inference

- is an argument that proceeds **from knowledge of a cause to a claim about an effect**, or, conversely, from knowledge of an effect to a claim about a cause.
    - **Example**:
- **From the knowledge that a bottle of wine had been accidentally left in the freezer overnight, someone might conclude that it had frozen (cause to effect).**
- **Conversely, after tasting a piece of chicken and finding it dry and crunchy, one might conclude that it had been overcooked (effect to cause).**

❑ Rely on Standard Terminology and Evaluating Arguments

- **Evaluating** argument means examining and justifying about **successfulness** and **failure of argument in general.**

- Once you understand the whatness of argument, you need to evaluate it.

- In any case, the **first and essential step** to understand an argument is to ***spot the conclusion*** ***or*** **find out the conclusion**.

Standard Terms for Evaluating DA and IA

- **Deductive Argument** can be evaluated by the following terms: **Valid, Invalid, Sound, and Unsound**

- **Inductive Argument** can be evaluated by the following terms: **Strong, Weak, Cogent, and Uncogent**

Three steps for evaluating arguments

1. Test for Deduction and Induction

*Do the Conclusion **necessarily** or **probably** follow from the premise?*

2. Test for Validity and Strength

Do the Premise really support the conclusion?

3. Test for soundness and Cogency

Are all premises true?

Examples of Valids
Cf: Sound vs Unsound

- All triangles have 3 sides
  Isosceles is a triangle
  Therefore, Isosceles has 3 sides (**Valid and Sound)**

- All Egyptians are Americans.
  All Americans are Ethiopians.
  Thus, All Egyptians are Ethiopians. (**Valid but Not sound)**

Cont….

- All sharks are birds
- All birds are politicians
- Thus, all Sharks are Politicians.(**Valid but Not sound)**

- All men are Ethiopians
- Obama is man
- Thus, Obama is Ethiopia.(**Valid but Not Sound**)

Examples of Invalids (Unsound)

- All Indians have long hair

  Dawit has Long hair

  Thus, Dawit is Indians.

- **All women are mammals.**

  Cat is mammals.

  Thus, cat is woman.

Cont….

- All philosophers are Scientist

S. Hawking is a scientist

Thus, Hawking is a Philosopher

- **All men are human**

Aster is human

Thus, Aster is man

Examples of Strong
Cf: Cogent vs Un Cogent

Example:

- **Identify the actual types of the following arguments.**

1. All previous Ethiopians PM were male. Hence, probably the next Ethiopian PM will be male.

2.From the basket contain 100 oranges, 5 oranges are selected randomly and were found to be ripe. Thus, probably all 100 oranges are ripe.

Cont.....

3. **If you live in Ethiopia, then you live in Africa.**

   <u>You Live in Africa</u>

   Therefore, you live in Ethiopia.

4. **Either Tomas is studying philosophy, or he is studying physics. <u>Tomas is not studying physics.</u>**

Therefore, Tomas is studying philosophy.

Cont…

5. **All emeralds previously found have been green.**

So, the next emerald to be found will be green.

6. **If the sum of 1+1=1 and 2+2 = 4, then it follows that the summation of 1+1 and 2+2 will be 5.**

Cont…

5. All emeralds previously found have been green.
So, the next emerald to be found will be green.

6. If the sum of 1+1=1 and 2+2 = 4, then it follows that the summation of 1+1 and 2+2 will be 5.

What is Good Argument?

What are the necessary condition to be a good argument?

- **A good Argument must meet *five (5) criteria***

❑ It must have the: -

1. Structural Principle:-A well-formed structure,
2. Relevance Principle:- Premises that are relevant to the truth of the conclusion,

3. Acceptability Principle- Premises that are acceptable to a reasonable person

4. Sufficiency Principle :-Premises that together constitute sufficient grounds for the truth of the conclusion, &

5. Rebuttal Principle:- Premises that provide an effective rebuttal to all anticipated criticisms of the argument

End!

Thank you